Corrections to Sections 6–8 of `notes.pdf` (DGLAP, running coupling, finite part)

Notation: $a \equiv \frac{k^+}{\vee}$, $c_0 = 2e^{-\gamma_E}$ (so $2\gamma_E + \log\frac{r^2\mu^2}{4} = \log\frac{r^2\mu^2}{c_0^2}$), $\beta_0 = \frac{11}{3}N_c$. Blue = new, red = removed.

1 Answers to your collaborator

1. **What are y and y' in the running coupling?** They are integration variables of $\frac{d^3N_{\rm LO}}{d^3k}$: the transverse positions at which the produced gluon crosses the target, y in the amplitude ($U(y)$) and y' in the complex-conjugate amplitude ($U(y')$). They come with the phase $e^{-ik\cdot(y'-y)}$, so $|y'-y| \sim 1/|\mathbf{k}|$. The scale of the coupling is $c_0/|y'-y|$, i.e. of order the gluon transverse momentum.
2. **Why are they outside the integral?** They should not be; that was a mistake in the notation. The bracket depends on $y'-y$, so it is a weight inside $\int_{x,y,x',y'}$. The same holds for the brackets in Sec. 6. Written correctly, Eqs. (2.1), (2.2) below put the weight f inside.
3. **The last term of DGLAP is wrong as written.**
   - In your $\frac{d^3N^y_{\xi\,\rm LO}}{d^3k}$ (8.3) nothing depends on y' except the phase. So $\int_{y'} e^{-ik\cdot(y'-y)/\xi} = (2\pi\xi)^2\delta^2(\mathbf{k})$, and (8.3) is *zero* for $\mathbf{k}\neq 0$. A log in front of it gives zero.
   - The log must be inside the y' integral. Then only $\log(y'-y)^2$ contributes, through $\int d^2y'\, e^{-ik\cdot(y'-y)/\xi}\log(y'-y)^2 = -\frac{4\pi\xi^2}{\mathbf{k}^2}$. The μ and the constant multiply $\delta^2(\mathbf{k})$ and drop out.
   - The term is therefore a number times H_4, Eq. (2.6). It has no pole and no μ, so it is not a DGLAP term: it is the subtraction of the small-ξ end of the $\frac{1-\xi}{\xi}$ part of $\mathcal{P}_{gg}$.

2 Definitions (weights inside the integrals)

$$\frac{d^3N_{\rm LO}}{d^3k}[f] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\, f(y'-y)\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\times\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x), \tag{2.1}$$

$$\frac{d^3N_{\xi\,\rm LO}}{d^3k}[f] \equiv \text{the same with } e^{-ik\cdot(y'-y)} \to e^{-ik\cdot(y'-y)\frac{1}{\xi}}, \tag{2.2}$$

$\frac{d^3N_{\rm LO}}{d^3k} \equiv \frac{d^3N_{\rm LO}}{d^3k}[1]$. Instead of (8.3), the y'-independent part of LO:

$$\frac{d^3N^y_{\rm LO}}{d^3k} \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x'}\frac{(x-y)\cdot(x'-y)}{(x-y)^2(x'-y)^2}\big[U^{e'c}(y)-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x), \tag{2.3}$$

$$H_4 = \frac{g^2N_c}{\pi\,\mathbf{k}^2}\frac{d^3N^y_{\rm LO}}{d^3k}, \tag{2.4}$$

with H_4 your term of (8.1), first line, without $-(1-\frac{k^+}{\vee})$ ($y \leftrightarrow z$; the w, w' integrals give $\frac{4\pi^2}{\mathbf{k}^2}$; checked in `g2iv_rc/compact/checks.py`). $\mathcal{P}_{gg}(\xi) = 2N_c\big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\big] + \frac{\beta_0}{2}\delta(1-\xi)$, with the standard plus distribution on $[0,1]$, used on $[a,1]$.

6 DGLAP (corrected)

Before renormalization (your (6.1), weights inside, last term with its y' integral done):

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{g^2}{8\pi^2}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,\frac{d^3N_{\xi\,\rm LO}}{d^3k}\Big[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big]$$
$$-\Big[\log\frac{\vee}{k^+}-1+\frac{k^+}{\vee}\Big]H_4 \;+\frac{g^2}{8\pi^2}\int\frac{d\xi}{\xi^2}2N_c\frac{1-\xi}{\xi}\log\big(\dots\big)\frac{d^3N^y_{\xi\rm LO}}{d^3k}. \tag{2.5}$$

The last term comes from

$$+\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\,2N_c\frac{1-\xi}{\xi}\Big(-\frac{4\pi\xi^2}{\mathbf{k}^2}\Big)\frac{d^3N^y_{\rm LO}}{d^3k}=-\frac{g^2N_c}{\pi\mathbf{k}^2}\frac{d^3N^y_{\rm LO}}{d^3k}\int_a^1 d\xi\,\frac{1-\xi}{\xi}=-\Big[\log\frac{1}{a}-(1-a)\Big]H_4. \tag{2.6}$$

After $\overline{MS}$ renormalization of $D_{h/g}$ (the pole $-\frac{g^2}{8\pi^2}\frac{2}{\epsilon}\,\mathcal{P}_{gg}\otimes\mathrm{LO}$ is absorbed, $\mu\to\mu_F$):

$$\boxed{\frac{d^3N_{\rm DGLAP}}{d^3k}=-\frac{g^2}{8\pi^2}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,\frac{d^3N_{\xi\,\rm LO}}{d^3k}\Big[\log\frac{(y'-y)^2\mu_F^2}{c_0^2}\Big]-\Big[\log\frac{\vee}{k^+}-1+\frac{k^+}{\vee}\Big]H_4} \tag{2.7}$$

- **The second term is not DGLAP** (no μ). It cancels the $+\log\frac{\vee}{k^+}H_4$ that the first term produces at its lower end, $\xi\to\frac{k^+}{\vee}$, from the $\frac{1-\xi}{\xi}$ part of $\mathcal{P}_{gg}$: there the phase forces $y'\to y$ and $\frac{d^3N_{\xi\rm LO}}{d^3k}[\log(y'-y)^2]\to-\frac{4\pi\xi^2}{\mathbf{k}^2}\frac{d^3N^y_{\rm LO}}{d^3k}$. That logarithm belongs to BFKL. So (2.7) as a whole contains no $\log\frac{\vee}{k^+}$; keep the two terms together.
- **Compact equivalent:** $-\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\Big\{\mathcal{P}_{gg}\frac{d^3N_{\xi\rm LO}}{d^3k}\big[\log\frac{(y'-y)^2\mu_F^2}{c_0^2}\big]-2N_c\frac{1-\xi}{\xi}\frac{d^3N^{\rm sub}_{\xi\rm LO}}{d^3k}\big[\log(y'-y)^2\big]\Big\}$. Here $\frac{d^3N^{\rm sub}_{\xi\rm LO}}{d^3k}[f]$ is (2.2) with $(x'-y')\to(x'-y)$ and $U(y')\to U(y)$ and the weight inside the y' integral. The integrand is then finite at $\xi\to0$.

7 Running coupling (corrected)

Before renormalization (your (7.1), weight inside), plus the $\overline{MS}$ counterterm of the coupling of LO ($\frac{d^3N_{\rm LO}}{d^3k}\propto g^2$):

$$\frac{d^3N_{\rm NLO}}{d^3k}=\frac{g^2\beta_0}{16\pi^2}\frac{d^3N_{\rm LO}}{d^3k}\Big[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big]\;\underbrace{-\frac{g^2\beta_0}{16\pi^2}\frac{2}{\epsilon}\frac{d^3N_{\rm LO}}{d^3k}}\;=\;\boxed{\frac{g^2\beta_0}{16\pi^2}\frac{d^3N_{\rm LO}}{d^3k}\Big[\log\frac{(y'-y)^2\mu_R^2}{c_0^2}\Big]} \tag{2.8}$$

Inside the integral this is the first term of the expansion of the running coupling at the scale $c_0/|y'-y|$:

$$\frac{d^3N_{\rm LO}}{d^3k}\Big|_{\alpha_s(\mu_R)}+(2.8)=\frac{1}{(2\pi)^3}\frac{2\,\alpha_s\big(c_0/|y'-y|\big)}{\pi\,k^+}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\times\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x)+O(\alpha_s^3), \tag{2.9}$$

with $\alpha_s(Q)=\alpha_s(\mu_R)\big[1-\frac{\alpha_s\beta_0}{4\pi}\log\frac{Q^2}{\mu_R^2}\big]$ and $\frac{g^2}{2\pi^2}=\frac{2\alpha_s}{\pi}$. The $\frac{\beta_0}{2}\delta(1-\xi)$ part of $\mathcal{P}_{gg}$ in (2.7) and (2.8) together leave $\frac{g^2\beta_0}{16\pi^2}\log\frac{\mu_R^2}{\mu_F^2}\frac{d^3N_{\rm LO}}{d^3k}$, which cancels the μ_R and μ_F dependence of LO and $D_{h/g}$.

8 Finite part (corrected only in the definitions)

$$\frac{d^3N_{\rm NLO}}{d^3k}=-\Big(1-\frac{k^+}{\vee}\Big)H_4$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}$$
$$\times\Big[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\Big]\big[\dots\big]\rho^{e'}(x')\rho^e(x) \tag{2.10}$$

is correct as you have it (H_4 as in (8.1)); remove (8.3) and use (2.1), (2.2), (2.4).

- **The H_4 terms.** If you add the last term of (2.7) here, the H_4 terms sum to $-\log\frac{\vee}{k^+}H_4$. That is not a new logarithm: it is cancelled by the lower end of the first term of (2.7). For this reason I recommend leaving it in (2.7).
- **At leading power** ($\frac{k^+}{\vee}\to0$): the lower limits become 0, $(1-\frac{k^+}{\vee})H_4\to H_4$, and the last term of (2.7) is $-\big[\log\frac{\vee}{k^+}-1\big]H_4$.